01529

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 2, Semester 1 (2025)

IT2020 – Software Engineering

Duration: 3 Hours

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Instructions to Candidates:

- This paper has 5 questions.
- Answer all questions in the booklet given.
- The total mark for the paper is 100.
- This paper contains 8 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1 (25 marks)

The National Health Services Derry (NHS-D) stands as a cornerstone of healthcare in Londonderry, Northern Ireland dedicated to delivering swift, compassionate, and dependable medical care to the community. Renowned for its commitment to excellence, NHS-D serves as a trusted lifeline for individuals and families, ensuring that every patient receives the timely and reliable support they deserve.

Analyze the partial requirements of the National Health Services Derry (NHS-D) given below and draw a **sequence diagram.**

Note:

- You may use suitable boundary, control, and entity classes.
- Processes for patient registration and login have already been implemented for you and are available under the modules titled "Register-NHS" and "Login-NHS".

Patients can easily register by providing essential personal details, including their full name, age, home address, contact number, and date of birth. Upon successful registration, each patient is assigned a unique NHS Identification Number. This identification number can be verified through the contact number provided during registration, ensuring both security and accurate identification within the system.

Once logged into the system, patients gain access to a user-friendly interface "ReserveUI" that allows them to search for medical consultants based on specific specializations. These specializations may include areas such as pediatrics, orthopedic surgery, chest medicine, and many others. After entering their desired specialization, the system will access "ResInfo" and display a list of available consultants along with their corresponding doctor codes. This feature helps patients quickly identify and connect with the appropriate healthcare professionals for their medical needs.

To book an appointment with a doctor, the patient must enter the doctor's code and check availability for the preferred date and time. The system will check "DocInfor" and get the doctor's availability and compare it with the patient's requested time slot. If the doctor is available, the patient will receive the availability status. Then they can confirm the appointment and proceed to the payment process. Then the system will get the payment confirmation from Payment Gateway. Once the payment is successful and completed, the system will generate an e-receipt for the patient and send it to the patient. If the doctor is unavailable at the selected date and time, the system will notify the patient of the unavailability and request alternative time slots.

Patients have the flexibility to book appointments with multiple doctors based on their individual healthcare needs and preferences. Whether consulting specialists across different fields or scheduling follow-up visits, the system supports booking with more than one doctor as required.

Question 2 (20 marks)

Given below is a detailed description of a web-based Hospital Management System developed for the government hospital network. Model a physical diagram for the following description.

Note: Use appropriate operating systems in necessary places.

This application can be used by both mobile and desktop/laptop users to manage their medical appointments and access their health records. Desktop users access the web application via HealthWeb_Client through a browser, while mobile users access it through HealthMob_App.

The core HealthCare_Service application resides inside the Apache Tomcat Web Server, which is deployed on an HP ProLiant DL380 server. HealthCare_Service consists of two main subcomponents: Appointment and Records. Both Desktop and Mobile devices interact with these through iAppointment and iRecords interfaces, respectively.

A patient can book appointments or view medical history using Patient_UI, built using ReactJS pages. Patient_UI resides inside the Nginx Webserver hosted on a Cisco UCS server. A patient can register to the system through desktop or mobile phone by accessing the iPatient interface of Patient_UI.

The backend Hospital_Management application, containing three main subcomponents: UserAccessControl, PatientDB, and Billing, is deployed on a Red Hat Enterprise Linux OS installed on a Dell EMC PowerEdge server. The iUserAccess interface provided by the Authentication component is accessed by both Appointment and Patient_UI components. The iPatientDB interface from the PatientDB component is utilized by the Records component via its iRecord interface. iPatientDB interface realized by PatientDB component is used by Billing component through its iBilling interface.

User.jar file which resides in the Hospital_Management application manifests the UserAccessControl component. Apache Tomcat and Nginx web servers communicate with Desktop device over HTTP protocol, while the Dell EMC PowerEdge server communicates with the HP ProLiant and Cisco UCS servers via a secure WAN connection.

Question 3 (25 marks)

Answer the questions given below using the partial code segment in Figure 1.

```
public class InventoryHandling {
    private int itemQty;

    public void setQtyLevel(int qty) {
        this.itemQty = qty;
    }

    public static void main(String[] args) {
        InventoryHandling item = new InventoryHandling();
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter reorder level(Low,Average,High)or-99 to exit:");
        String input = scanner.nextLine();

        while (!input.equals("-99")) {

             switch (input) {
                case "Low":
                    item.setQtyLevel(50);
                    System.out.println("Reorder Level updated to 50!");
                    break;

                case "Average":
                    item.setQtyLevel(20);
                    System.out.println("Reorder Level updated to 20!");
                    break;

                case "High":
                    item.setQtyLevel(10);
                    System.out.println("Reorder Level updated to 10!");
                    break;

                default:
                    System.out.println("Invalid input!");
            }
       System.out.print("Enter reorder level(Low,Average,High)or-99 to exit:");
       String input = scanner.nextLine();
        }

        scanner.close();
    }
}
```

Figure 1: Partial coding segment

a) Calculate the statement coverage for the code shown in Figure 1, when the user enters the input values 'Low', 'High', and '-99' in sequence. (06 marks)

b) What is the statement coverage if the user enters '-99' as the reorder level on the first attempt? (05 marks)

c) Draw a control flow graph for the given code segment and identify the total number of branches present in the code. (04 marks)

d) Calculate the branch coverage when the following inputs are entered in sequence: 'Low', 'Average', 'Medium', and '-99' (05 marks)

e) Calculate the branch coverage when the following inputs are entered in sequence: 'Max' and '-99' (05 marks)

Question 4 (20 marks)

a)

In a video game development system, different types of characters are available. Some characters have basic abilities, such as moving and attacking, while others have additional features such as invisibility, teleportation, or the ability to fly. Each ability can be added dynamically to a character, but the system must be able to add or remove these abilities without changing the original character.

The game developers want to ensure that the game system can handle new abilities that might be introduced in the future, while also maintaining the flexibility to update a character's abilities without modifying the existing code for each ability.

For example, a character may start with basic movement and attacking abilities, and later, the developer may want to add the ability to teleport or become invisible without having to create entirely new subclasses for each combination of abilities.

i) Recommend the most suitable design pattern for the given scenario above and justify your recommendation. (02 marks)

ii) Draw the class structure of the design pattern that you identified above in part i) with appropriate classes and methods for the given scenario. (05 marks)

b)

In a stock market monitoring system, multiple investors are interested in tracking the prices of different stocks. Whenever the price of a stock changes, all investors who have subscribed to updates for that stock should be notified immediately. The system should allow investors to subscribe or unsubscribe from updates dynamically.

i) Recommend the most suitable design pattern for the given scenario above and justify your recommendation. (02 marks)

ii) Draw the class structure of the design pattern that you identified above in part i) with appropriate classes and methods for the above scenario. (05 marks)

c) Consider the code segment in Figure 2 and answer the questions provided at the end.

```
class Engine {
    public void start() {
        System.out.println("Engine started");
    }
}

class AirConditioner {
    public void turnOn() {
        System.out.println("Air Conditioner turned on");
    }
}

class MusicSystem {
    public void playMusic() {
        System.out.println("Music system is playing");
    }
}


class CarController {
    private Engine engine;
    private AirConditioner ac;
    private MusicSystem musicSystem;

    public CarController() {
        engine = new Engine();
        ac = new AirConditioner();
        musicSystem = new MusicSystem();
    }

    public void startCar() {
        engine.start();
        ac.turnOn();
        musicSystem.playMusic();
        System.out.println("Car is ready to drive!");
    }
}
```

Figure 2: "Car Startup Control System" Code segment

i) Identify the design pattern used in the code in Figure 2. (01 mark)

ii) Justify your answer briefly, referring to how the structure and behavior of the code relate to the chosen design pattern. (01 mark)

iii) Draw the class diagram that represents the structure of the above implementation. (04 marks)

Question 5 (10 marks)

a) What is the difference between "black-box Testing" and "White-box Testing"? (02 marks)

b) Differentiate between centralized and distributed version control systems. (03 marks)

c) What is the purpose of a change request form in change management? Mention two elements it typically includes. (03 marks)

d) Define configuration management and list any two activities involved. (02 marks)

---------- END OF THE QUESTION PAPER ----------